

ICA at the Cocktail Party: Casting Instrument-free Omitted Variable Bias Correction as a Blind Source Separation Problem

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July 22, 2025

Abstract

Endogeneity due to omitted variables poses a persistent challenge in empirical research, particularly in marketing contexts where valid instruments are difficult to find. While instrument-free approaches such as the Gaussian copula control function (GCCF) have gained popularity, recent evidence shows that GCCF fails in the traditional omitted variable bias case, often producing more biased estimates than OLS. This paper proposes a novel instrument-free method tailored to this prevalent form of endogeneity. Building on minimal assumptions (the sum of omitted variables is normally distributed and affects the endogenous regressor additively, and the existence of non-normal exogenous noise in the endogenous regressor) our approach uses Independent Component Analysis (ICA) to disentangle the unobserved normal confounder from the exogenous, non-normal variation in the regressor. The recovered confounder is then incorporated as a control function, enabling consistent estimation of causal effects without external instruments. We prove formal identification under the proposed data-generating process and demonstrate through simulations that the method delivers unbiased estimates bias. An R code implementation is provided.

Keywords Endogeneity, omitted variable bias, instrument-free methods, independent component analysis (ICA)

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1 Introduction

In empirical research, a key challenge is that regressors are often endogenous, that is correlated with the error term, due to omitted variables. The classical remedy uses instrumental variables (IVs) to correct for this bias, but finding valid instruments that satisfy the exclusion restriction is notoriously difficult. In response, a range of instrument-free approaches leverage distributional assumptions to achieve identification without observed instruments. Examples include using higher moments or heteroskedasticity to generate instruments (Lewbel, 1997; Rigobon, 2003) and the latent instrumental variables (LIV) approach (Ebbes et al., 2005) which introduces a latent variable to account for regressor–error dependence. These methods exploit features like non-normality or variance patterns in the data to effectively create or uncover assumed “instruments” from the data itself. In many marketing contexts such methods offer an appealing alternative for causal inference.

One instrument-free method that has gained wide adoption in marketing is the Gaussian copula control function (GCCF) approach introduced by Park and Gupta (2012). The appeal of this method lies in its seemingly minimal and intuitive assumptions. In essence, it assumes: (1) the endogenous regressor is non-normally distributed; (2) the error term is normally distributed; and (3) the dependence between the regressor and error follows a Gaussian copula (i.e. they form a joint normal pair). Assumption (1) is testable (researchers can check if the regressor’s empirical distribution deviates from normality) and in practice this often holds, since many marketing variables (prices, sales, etc.) exhibit skew or other non-normal features. Assumptions (2) and (3) are not directly testable, but they are viewed as plausible: normal errors are a standard modeling assumption in linear models, and Gaussian correlation provides a flexible way to capture the regressor–error relationship. Crucially, if a dataset satisfies these conditions, the copula method enables a consistent estimation of the regression parameters without an actual instrument. This convenience has made the Gaussian copula approach popular in empirical studies addressing endogeneity. Researchers embraced

it as a general solution that “directly addresses the issue of regressor–error correlation by assuming a plausible relationship between the endogenous regressor and the error” (Park and Gupta, 2024), thereby avoiding the hunt for elusive instruments.

Despite its popularity, recent evidence suggests that researchers’ trust in the Gaussian copula control function method may not be warranted. In particular, Dost and Haschka (2025) uncover that the method’s validity hinges on additional, often unrecognized assumptions, and they demonstrate that GCCF fails to solve the most common form of endogeneity: a linear additive omitted variable bias. In a classic omitted confounder scenario, where an unobserved factor adds both to the independent variable and to the outcome, the Gaussian copula approach breaks down and can turn out even more biased than ordinary OLS. The intuition is that the Gaussian copula method assumes a particular cause for non-normality of the endogeneous variable (a bijective function transforming the otherwise joint-normal dependence with the error term). If the true endogeneity stems from a straightforward omitted variable (e.g., a missing factor that linearly contributes to both X and Y), this situation may violate the method’s underlying “bijective transform” condition and cannot be adequately captured by a Gaussian copula. Unfortunately, such omitted variable bias is the most prevalent endogeneity problem in marketing applications, as unobserved factors (e.g., consumer tastes, competitive actions) often correlate with firms’ decisions. Thus, paradoxically, the GCCF method falls short exactly in the scenario that researchers most frequently confront. These findings have cast doubt on instrument-free copula corrections and highlight the need for new solutions that can handle traditional confounding structures.

This research proposes a novel instrument-free method specifically designed for the ubiquitous case of additive omitted variable bias. Our approach retains the spirit of minimal assumptions, but shifts the perspective on the data-generating process. We assume the following about the true data-generating process (DGP): (1) the structural error term is normally distributed (an untestable but standard assumption in linear regression models); (2) the key omitted confounding variable is also normally distributed; and (3) the endogenous

regressor is non-normally distributed. Assumption (3) is testable (one can verify that the regressor’s distribution is not Gaussian). The critical implicit assumption behind it is that the source of the non-normality is some independent, exogenous variation in the regressor – in other words, the part of P that makes its distribution non-normal comes from an outside factor unrelated to the outcome except through X . These assumptions closely mirror the classic IV scenario: we essentially posit that there exists a latent instrument, an unobserved exogenous factor that influences P (introducing non-normal variation) but does not directly affect Y . At the same time, a latent normally distributed confounder drives the endogeneity by affecting both X and Y . This setup is analogous to the LIV approach of Ebbes et al. (2005), which models a discrete latent instrument to achieve identification. However, in our approach the exogenous variation in X can follow any non-normal distribution (continuous or discrete), granting more flexibility. By not restricting the latent instrument to a particular form (e.g., a fixed number of mass points as in a discrete latent class model), the method can exploit whatever non-normality is present in X ’s distribution, which is expected to improve efficiency in practice.

Methodologically, we reconceptualize the endogeneity problem as a blind source separation task. This is inspired by the classic “cocktail party problem,” where one aims to recover original independent signals from observed mixtures of those signals. Here, the two “signals” are (i) the unobserved normal confounder and (ii) the unobserved non-normal exogenous component of X . The observed endogenous regressor P and the dependent variable Y are each mixtures of these two underlying signals: intuitively, P is driven by both the confounder and the exogenous input, while Y is driven by both the confounder and the effect of X (plus random noise). If these latent factors are statistically independent and at least one has a non-Gaussian distribution, we can separate them using Independent Component Analysis (ICA). ICA is a computational technique originally developed to solve exactly such signal-separation problems by exploiting non-Gaussianity (Comon, 1994). In our context, applying ICA to the joint data on P and Y allows us to disentangle the two independent

components: one component that is Gaussian (which we interpret as the confounding latent variable), and another that is non-Gaussian (the exogenous variation in P). Once we have isolated the latent normal confounder, we can incorporate it into the outcome regression as a control function to absorb the endogeneity. In effect, we recover what an ideal instrument or control variable would have captured: the part of Y 's variation due to the confounder. Adding it to the estimated model as a control function removes its confounding, leaving a consistent estimate of the true effect of P on Y . We show formally that under the stated assumptions, the model is point identified: the structural parameter (e.g. the causal effect of P) can be uniquely recovered. In summary, our approach leverages a novel combination of distributional assumptions and signal-processing algorithms (ICA) to solve a traditional econometric problem without requiring any external instruments.

This work makes several contributions to the literature on endogeneity and marketing analytics:

- Minimal, plausible assumptions: The proposed method operates under just three intuitive assumptions – normality of the error and confounder, and non-normality of the regressor – which are plausible in many marketing settings. Normal errors and latent factors are routinely assumed in econometric models, and by the Central Limit Theorem a combination of multiple unobserved influences tends toward normal (justifying the normal confounder assumption). Meanwhile, many marketing endogenous variables (e.g. advertising expenditures, prices, etc.) are empirically non-normal, providing the needed leverage for our method. These are comparable or even weaker conditions than those behind existing instrument-free methods like Gaussian copula or LIV.
- Addressing the most common endogeneity form: Our method directly targets omitted variable bias in an additive model, arguably the most common form of endogeneity in marketing research. Unlike the Gaussian copula approach, which fails under a classic confounder scenario, the proposed technique is specifically designed for it. It effectively

reconstructs the impact of the unobserved confounder and removes it from the regression bias. This fills an important gap for marketing researchers who need to correct for latent confounders but lack good instruments.

- Improved accuracy and robustness: Through extensive simulation studies, we demonstrate that the new method produces consistent and unbiased estimates across a variety of data-generating processes. It performs favorably against existing alternatives, often yielding lower bias and variance (higher efficiency) than both the Gaussian copula correction and the LIV estimator, especially in the presence of a strong omitted variable. The approach is also shown to be robust to moderate mis-specifications (for instance, if the latent confounder is not perfectly normal or there are multiple confounders) – the estimator still delivers substantial bias reduction in such cases. These results suggest that the method not only addresses endogeneity effectively but does so with practical reliability under realistic conditions.
- Practical implementation: We provide an accessible implementation of the method for researchers and practitioners. In particular, we supply an open-source R code function that automates the key steps of the procedure – from performing the ICA-based signal separation, to generating the control function for inclusion in regression, to providing robust standard errors with an adapted bootstrap.¹ The implementation can seamlessly accommodate additional observed control variables alongside the latent one, and includes diagnostics to guide the user (for example, checking the non-normality requirement). By lowering the technical barriers, we aim to make instrument-free endogeneity correction using our method as convenient as running a two-stage regression.

To summarize, our proposed approach contributes a theoretically sound and empirically effective solution for endogeneity correction from omitted variables in the absence of instruments, tailored to the needs of marketing researchers. The remainder of the paper is

¹ A user-friendly R function is available on <https://github.com/HashtagHaschka/ICA-based-endogeneity-correction>.

organized as follows. First, we further elaborate the endogeneity problem and the conceptual foundation of our ICA-based solution. Next, we develop the formal identification proof and estimation procedure under the assumed DGP. Then, we present a series of simulation studies that benchmark the new method’s performance against existing approaches across various scenarios. Finally, we discuss the implications of our findings for researchers and managers, outline the method’s limitations, and suggest future research directions.

2 Endogeneity from omitted variables

Consider a structural linear model

$$Y = \alpha + \beta_1 P + \xi, \quad (1)$$

where P is a scalar endogenous regressor correlated with the error term ξ . Let us further assume that the endogenous regressor can, and the error might be decomposed as:

$$P = \eta + \rho v \quad \text{and} \quad \xi = v (+u), \quad (2)$$

where v is a normally distributed component with zero mean, i.e., $v \sim \mathcal{N}(\cdot)$. In (2), we allow the error to consist of an exogenous component, i.e., $u \perp (v, \eta)$, without specifying its distribution. For identification with the proposed method, P must consist of a non-normal exogenous component η , i.e., $\eta \not\sim \mathcal{N}(\cdot)$ and $\eta \perp (v, u)$.²

Because P and ξ share a common normal component, P is endogenous. Estimation with OLS will yield biased estimates for β_1 . To correct for the bias requires to separate the non-normal exogenous from the normal endogenous variance in P , or to extract the common normal endogenous component in P and ξ . We propose using Independent Component

² Specifically, identification is based on the assumption that higher cumulants of η and ξ differ. There exist no distribution with higher cumulants equal to the normal distribution, but different lower cumulants.

Analysis (ICA) to do so.

2.1 Intuition for ICA-based omitted component extraction

Independent Component Analysis is a statistical technique that seeks to recover latent independent factors from observed multivariate data. In essence, ICA assumes that each observed variable is a linear mixture of some underlying source components, and it aims to unmix the data to find those original sources. Unlike traditional principal component analysis (PCA) or factor analysis which rely only on second-order (covariance) structure, ICA exploits higher-order statistics to detect structures that remain “hidden” to covariance-based methods (Nordhausen and Oja, 2018). In practice, this means ICA looks for a linear transformation of the data that produces new variables (components) which are as statistically independent from each other as possible (Comon, 1994; Hyvärinen and Oja, 2000). These independent components are typically assumed to be non-Gaussian in distribution, since non-Gaussianity is key to identifying truly independent factors (Hyvärinen and Oja, 2000). By maximizing independence (or equivalently minimizing mutual information) between components, ICA effectively separates the distinct underlying influences in the data.

ICA is widely regarded as a form of “blind source separation” – “blind” because it makes minimal assumptions about the mixing process or the nature of the sources, and only uses the data itself to recover the signals. Classic illustrations include the “cocktail party problem”, where ICA can isolate individual speakers’ voices from a jumble of simultaneous recordings. In an econometric context, ICA can be seen as a latent variable technique that generalizes factor analysis by not requiring the latent factors to be uncorrelated Gaussian; instead, they can be non-Gaussian and truly independent (Nordhausen and Oja, 2018).

In our regression problem, where we suspect an omitted variable v influences both the predictor P and the error term ξ (thus confounding the estimate of effect β_1), ICA offers a compelling solution. We can treat the measured variables (for instance, the observed endogenous variable P and the dependent variable Y) as mixtures of the true independent

influences: one latent component is the omitted factor v , and the other is the exogenous component η in P (assumed non-normal) and any idiosyncratic noise u in Y being added to either v (if also normal), or η (if non normal). ICA will exploit the statistical independence and non-normality assumptions to unmix these signals. In intuitive terms, it will pull apart the tangled contributions of v vs. η (and u), yielding separate estimates of the hidden variable and the genuine predictor. By extracting the influence of v from both P and ξ , we can then correct the regression – for example, by including the recovered v as a control, similar to a control function method. Thus, ICA allows us to address the omitted variable problem by recovering (or removing) the unobserved factor’s effect, isolating the true causal relationship between P and Y .

One well-known implementation of ICA is the JADE algorithm, introduced by Cardoso and Souloumiac (1993). JADE stands for Joint Approximate Diagonalization of Eigenmatrices, reflecting its core approach: JADE uses higher-order statistics, in particular, fourth-order cumulants, which relate to measures like kurtosis, to identify the independent components. The key insight is that if the data are correctly “unmixed” into independent sources, certain cross-cumulants (analogs of covariance that involve fourth powers of the data) will vanish for distinct components. JADE computes several such fourth-order cumulant matrices of the observed data and then finds a single linear transform (a rotation matrix) that simultaneously makes all these matrices as diagonal as possible. Diagonalization here means each output component has no cross-correlation at the fourth-order with any other – a signature of statistical independence (Cardoso and Souloumiac, 1993).

In our context of an omitted variable, the JADE algorithm would help by leveraging the non-normality of one component (the η in P) versus the normality of the omitted v . Suppose $P = v + \eta$ (with η non-Gaussian) and $\xi = v + u$ (with u independent noise). JADE would analyze the joint data (here: P and Y) and use fourth-order moment structure to separate two independent signals. In effect, it can pull out the non-Gaussian source η as one independent component, and isolate the remaining (approximately Gaussian) combination

that contains v and u (if u is normal) as a second component. By doing so, JADE has indirectly identified the presence of v : even if v itself is entangled with u , JADE’s separation reveals that one of the outputs is a shared factor influencing both original variables. We could interpret that output as “the omitted factor (plus any Gaussian noise)” and thereby adjust our regression accordingly. In practical terms, using JADE on the data would allow us to remove the influence of v from P , or to include an estimated v variable in the model, thus correcting the bias (we propose the latter). The strength of JADE here is that it provides a clear, matrix-based solution to find what combination of P and Y (or P and ξ) corresponds to an independent hidden factor. By finding that hidden factor, ICA could offer a data-driven way to account for an omitted confounder. Using the hidden factor in the regression allows estimating the regression coefficient β_1 without the distortion that came from v , improving the causal interpretability of the model.

2.2 Proof of identification

The following proof draws on previous work by Jiang (2025). Reconsider the model in (1)-(2), but rewrite it as a simultaneous equations system:

$$P = \eta + \rho\xi \tag{3}$$

$$Y = \alpha + \beta_1 P + \xi \tag{4}$$

where:

- η is non-normal with $\kappa_3(\eta) \neq 0$ **or** $\kappa_4(\eta) \neq 0$
- $\xi \sim \mathcal{N}(0, \sigma^2)$ with $\kappa_3(\xi) = \kappa_4(\xi) = 0$
- $\eta \perp\!\!\!\perp \xi$ (mutually independent)

The observed vector is $\mathbf{X} = [P, Y]^\top$. Rewrite in matrix form:

$$\mathbf{X} = \mathbf{A}\mathbf{S}, \quad \mathbf{S} = \begin{bmatrix} \eta \\ \xi \end{bmatrix}, \quad \mathbf{A} = \begin{bmatrix} 1 & \rho \\ \beta_1 & 1 + \rho\beta_1 \end{bmatrix} \quad (5)$$

The structural parameter matrix is $\mathbf{\Lambda} = \mathbf{A}^{-1} = \begin{bmatrix} 1 + \rho\beta_1 & -\rho \\ -\beta_1 & 1 \end{bmatrix}$.

Cumulant assumptions

Assumption 1 (Invertibility (A1)). $\det(\mathbf{A}) = 1 \cdot (1 + \rho\beta_1) - \rho\beta_1 = 1 \neq 0$, so $\mathbf{A}$ is invertible (Assumption A1 Jiang, 2025).

Assumption 2 (Non-zero h -th cumulant (A2)). There exists $h \in \{3, 4\}$ such that $\kappa_h(\eta) \neq 0$ and $\kappa_h(\xi) = 0$ (Assumption A2 Jiang, 2025).

Assumption 3 (Diagonal h -th cumulant tensor (A3)). The h -th cumulant tensor $C_h(\mathbf{S})$ is diagonal (holds by independence of η and ξ) (Assumption A3 Jiang, 2025).

Cumulant objective function

For the chosen $h \in \{3, 4\}$, define:

$$Q_h(\mathbf{w}) = \kappa_h(\mathbf{w}^\top \mathbf{X}) = \begin{cases} \mathbb{E}[(\mathbf{w}^\top \mathbf{X})^3] & \text{if } h = 3 \\ \mathbb{E}[(\mathbf{w}^\top \mathbf{X})^4] - 3(\mathbb{E}[(\mathbf{w}^\top \mathbf{X})^2])^2 & \text{if } h = 4 \end{cases} \quad (6)$$

Lemma 1 (Hessian factorization for h). Under Assumptions 1-3, the Hessian satisfies:

$$\nabla_{\mathbf{w}}^2 Q_h(\mathbf{w}) = \mathbf{A} \mathbf{D}(\mathbf{w}) \mathbf{A}^\top \quad (7)$$

where $\mathbf{D}(\mathbf{w}) = \text{diag}(d_1(\mathbf{w}), d_2(\mathbf{w}))$ with

$$d_1(\mathbf{w}) = h(h-1)(\mathbf{w}^\top \mathbf{a}_1)^{h-2} \kappa_h(\eta), \quad d_2(\mathbf{w}) = 0 \text{ because } \kappa_3(\epsilon) = 0 \text{ for } h \geq 3. \quad (8)$$

Proof. With $\mathbf{v} = \mathbf{A}^\top \mathbf{w}$, the h -th cumulant decomposes as:

$$\kappa_h(\mathbf{v}^\top \mathbf{S}) = v_1^h \kappa_h(\eta) + v_2^h \kappa_h(\xi) = (\mathbf{w}^\top \mathbf{a}_1)^h \kappa_h(\eta), \quad (9)$$

since $\kappa_h(\xi) = 0$. The Hessian with respect to $\mathbf{v}$ is:

$$\nabla_{\mathbf{v}}^2 \kappa_h(\mathbf{v}^\top \mathbf{S}) = \text{diag}(h(h-1)v_1^{h-2} \kappa_h(\eta), 0). \quad (10)$$

By the chain rule ($\mathbf{v} = \mathbf{A}^\top \mathbf{w}$):

$$\nabla_{\mathbf{w}}^2 Q_h(\mathbf{w}) = \mathbf{A} (\nabla_{\mathbf{v}}^2 \kappa_h(\mathbf{v}^\top \mathbf{S})) \mathbf{A}^\top = \mathbf{A} \mathbf{D}(\mathbf{w}) \mathbf{A}^\top. \quad \square \quad (11)$$

□

Eigenvector recovery

Theorem 1 (Eigenvector recovery for h). *For $\mathbf{w}^{(1)} \neq \mathbf{w}^{(2)} \in \mathbb{R}^2$ with $\mathbf{w}^{(k)\top} \mathbf{a}_1 \neq 0$ ($k = 1, 2$), such that $d_1(\mathbf{w}^{(k)}) \neq 0$, define:*

$$\mathbf{H} = [\nabla^2 Q_h(\mathbf{w}^{(2)})]^+ \nabla^2 Q_h(\mathbf{w}^{(1)}) \quad (12)$$

where $^+$ denotes Moore-Penrose pseudoinverse. Then:

$$\mathbf{H}^\top = \mathbf{A} \mathbf{M} \mathbf{A}^{-1}, \quad \mathbf{M} = \text{diag}(m, 0), \quad m = \left(\frac{\mathbf{w}^{(1)\top} \mathbf{a}_1}{\mathbf{w}^{(2)\top} \mathbf{a}_1} \right)^{h-2}. \quad (13)$$

The first column of $\mathbf{A}$ (i.e., $\mathbf{a}_1 = [1, \beta_1]^\top$) is the eigenvector of $\mathbf{H}^\top$ for eigenvalue $m \neq 0$ (Jiang, 2025, Theorem 5.1).

Proof. From Lemma 1, $\nabla^2 Q_h(\mathbf{w}) = \mathbf{A} \mathbf{D}(\mathbf{w}) \mathbf{A}^\top$ with $\mathbf{D}(\mathbf{w}) = \text{diag}(d_1(\mathbf{w}), 0)$. The pseudoinverse is:

$$[\nabla^2 Q_h(\mathbf{w}^{(2)})]^+ = (\mathbf{A}^\top)^{-1} \mathbf{D}(\mathbf{w}^{(2)})^+ \mathbf{A}^{-1}, \quad \mathbf{D}(\mathbf{w}^{(2)})^+ = \text{diag}(d_1(\mathbf{w}^{(2)})^{-1}, 0). \quad (14)$$

Substituting yields:

$$\mathbf{H} = (\mathbf{A}^\top)^{-1} \mathbf{D}(\mathbf{w}^{(2)})^+ \mathbf{D}(\mathbf{w}^{(1)}) \mathbf{A}^\top \quad (15)$$

$$= (\mathbf{A}^\top)^{-1} \text{diag}\left(\frac{d_1(\mathbf{w}^{(1)})}{d_1(\mathbf{w}^{(2)})}, 0\right) \mathbf{A}^\top. \quad (16)$$

Transposing gives:

$$\mathbf{H}^\top = \mathbf{A} \text{diag}\left(\frac{d_1(\mathbf{w}^{(1)})}{d_1(\mathbf{w}^{(2)})}, 0\right) \mathbf{A}^{-1} \quad (17)$$

$$= \mathbf{A} \mathbf{M} \mathbf{A}^{-1}, \quad (18)$$

where $\frac{d_1(\mathbf{w}^{(1)})}{d_1(\mathbf{w}^{(2)})} = \left(\frac{\mathbf{w}^{(1)\top} \mathbf{a}_1}{\mathbf{w}^{(2)\top} \mathbf{a}_1}\right)^{h-2}$. Eigendecomposition confirms:

$$\mathbf{H}^\top \mathbf{a}_1 = \mathbf{A} \mathbf{M} \mathbf{A}^{-1} \mathbf{a}_1 = \mathbf{A} \mathbf{M} \mathbf{e}_1 = m \mathbf{a}_1. \quad \square \quad (19)$$

□

Remark 1 (One zero-cumulant component (pseudoinverse)). When one latent source (ξ) has zero h -th cumulant, $\nabla^2 Q_h(\mathbf{w})$ is rank-deficient. For $h = 3$:

$$\nabla^2 Q_h(\mathbf{w}) = \mathbf{A} \text{diag}(d_1(\mathbf{w}), 0) \mathbf{A}^\top$$

with $d_1(\mathbf{w}) = 6(\mathbf{w}^\top \mathbf{a}_1)\kappa_3(\eta)$. Its Moore-Penrose inverse is:

$$[\nabla^2 Q_h(\mathbf{w})]^+ = (\mathbf{A}^\top)^{-1} \text{diag}(d_1(\mathbf{w})^{-1}, 0) \mathbf{A}^{-1},$$

which is exactly used in Theorem 1. The same logic applies for $h = 4$ with $d_1(\mathbf{w}) = 12(\mathbf{w}^\top \mathbf{a}_1)^2 \kappa_4(\eta)$. See general discussion in Jiang (2025, Section 5.2).

Remark 2 (Nonsquare mixing (factor models)). If $\mathbf{X} \in \mathbb{R}^p$, $\mathbf{S} \in \mathbb{R}^d$ with $p > d$ and $\mathbf{A}$ is $p \times d$, the cumulant strategy extends. The Hessian becomes:

$$\nabla^2 Q_h(\mathbf{w}) = \mathbf{A} \text{diag}(d_1(\mathbf{w}), \dots, d_d(\mathbf{w})) \mathbf{A}^\top,$$

a $p \times p$ matrix of rank $\leq d$. Using Moore-Penrose inverses and forming $\mathbf{H}$ as in Theorem 1 recovers columns of $\mathbf{A}$ associated with non-zero cumulants. See Jiang (2025, Section 6.1, Equations 6.3-6.5).

Point identification

Theorem 2 (Identification). Under Assumptions 1-3, β_1 is point identified (Jiang, 2025, Theorem 5.2).

Proof. 1. **Recover $\mathbf{a}_1$:** Compute $\mathbf{H}$ for $\mathbf{w}^{(1)}, \mathbf{w}^{(2)}$ with $\mathbf{w}^{(k)\top} \mathbf{a}_1 \neq 0$. The eigenvector of $\mathbf{H}^\top$ for the non-zero eigenvalue satisfies $\mathbf{v} \propto \mathbf{a}_1 = [1, \beta_1]^\top$.

2. **Normalize:** With $\mathbf{v} = [v_P, v_Y]^\top$ and $v_P \neq 0$,

$$\beta_1 = \frac{v_Y}{v_P}. \quad \square \tag{20}$$

□

Corollary 1. If ξ is normal and η has either $\kappa_3(\eta) \neq 0$ or $\kappa_4(\eta) \neq 0$, then β_1 is identified without instruments or zero-correlation assumptions (Jiang, 2025, Corollary 1).

Implementation remarks

Remark 3 (Role of h). *The identification strategy works for either:*

- $h = 3$: Requires $\kappa_3(\eta) \neq 0$ (eigenvalue $m = \frac{\mathbf{w}^{(1)\top} \mathbf{a}_1}{\mathbf{w}^{(2)\top} \mathbf{a}_1}$)
- $h = 4$: Requires $\kappa_4(\eta) \neq 0$ (eigenvalue $m = \left(\frac{\mathbf{w}^{(1)\top} \mathbf{a}_1}{\mathbf{w}^{(2)\top} \mathbf{a}_1} \right)^2$)

The eigenvector $\mathbf{a}_1$ is identical in both cases. For $h = 4$, the eigenvalue is squared but the eigenvector remains unchanged. Lewbel et al. (2024) consider the $h = 3$ case, Jiang (2025) provide extension to $h = 4$. Both rely on the mix of one non-Gaussian and one Gaussian latent, and extract slopes via ratios of higher-order moments (or equivalently, eigenvectors of Hessians).

Remark 4 (Finite-sample considerations). *While Assumptions 1-3 suffice for theoretical identification of β_1 , in finite-sample estimation one must consider three practical issues:*

- **Sample size.** *Sample estimators require $n \geq \binom{d+h-1}{h}$ for h -th cumulant estimation*
- **Empirical noise.** *When $\text{Var}(\eta)$ is small relative to $\text{Var}(\xi)$, the sample third or forth cumulant $\hat{\kappa}_3(\eta)$ and $\hat{\kappa}_4(\eta)$ can be dominated by sampling variation, making $\nabla^2 \hat{Q}(w)$ numerically unstable.*
- **Distributional smoothness.** *A continuous, non-Gaussian distribution for η helps ensure that its third and forth cumulants are well-behaved and that higher-order sample moments converge at the usual rate.*

These concerns affect only the accuracy and stability of the estimator in finite samples, not the underlying algebraic identification, which relies solely on $\kappa_h(\eta) \neq 0$ and independence.

Remark 5 (Inclusion of an exogenous regressor). *Suppose the structural model includes an additional observed exogenous regressor X :*

$$P = \eta + \rho \xi + \tau X, \quad Y = \alpha + \beta_1 P + \beta_2 X + \xi,$$

with $X \perp\!\!\!\perp \xi$. In this case one proceeds by partialling out X :

1. Regress P on X and form residuals $P^\perp = P - \hat{\tau} X$.
2. Regress Y on X and form residuals $Y^\perp = Y - \hat{\beta}_2 X$.

Then $(P^\perp, Y^\perp)$ satisfy exactly the same system as in equations (3)–(4), but without the X term. All cumulant assumptions 1–3 continue to hold on the residualised variables, and the Hessian factorization, pseudoinverse step, and eigenvector-ratio identification carry through verbatim upon replacing $P \mapsto P^\perp$ and $Y \mapsto Y^\perp$.

3 Implementation with example code

In the following, we present the example code and outline how the proposed approach can be implemented in the R software. Since the approach has a simple correction function representation, interested users can easily replicate it themselves, as it is only slightly more complicated than an IV estimator. Nevertheless, we also provide a user-friendly R function on GitHub with example codes and applications on real data sets.³

3.1 One endogenous regressor

1. Use ICA to decompose the observed data (dependent variable Y and endogenous regressor P) into two statistically independent components.
2. Identify the component that is the most normal.
3. Regress Y on P and the identified normal component from ICA. This step mimics a control function or instrumental variable logic by purging the endogeneity via the ICA decomposition.

³ <https://github.com/HashtagHaschka/ICA-based-endogeneity-correction>


```

1 # Step 1: Set simulation parameters
2 N <- 1000                # number of observations
3 rho <- 0.8               # strength of endogeneity
4 alpha <- 2               # intercept
5 beta <- 1                # slope coefficient
6 # Step 2: Create data
7 # Random noise (e.g., omitted factors)
8 xi <- rnorm(N)
9 # Exogenous influence (could be thought of as an instrument)
10 eta <- rgamma(N, shape = 1)
11 # Endogenous variable: combines error and exogenous component
12 P <- rho * xi + eta
13 # Outcome variable: depends on P and the error
14 Y <- alpha + beta * P + xi
15 # Step 3: Apply Independent Component Analysis (ICA)
16 data_matrix <- cbind(Y, P)
17 ica_result <- ica(X = data_matrix, nc = 2, method = "jade")
18 # Step 4: Find the ICA component that looks most normally
    distributed
19 hist(ica_result$S[, 1])
20 hist(ica_result$S[, 2])
21 # Step 5: Estimate the regression model, controlling for the normal
    component
22 model <- lm(Y ~ P + ica_result$S[, 1])
23 # View the results
24 summary(model)

```

Listing 1: R code for simulation and estimation using ICA to address endogeneity.

3.2 One endogenous and one exogenous regressor

If there is an exogenous regressor in the model, we first eliminate all (linear) dependence with the endogenous regressor and the dependent variable.

1. Regress endogenous regressor P on exogenous regressor X and obtain the residuals $P^\perp$.
2. Regress dependent variable Y on exogenous regressor X and obtain the residuals $Y^\perp$.
3. Use ICA to decompose these residuals (dependent variable $Y^\perp$ and endogenous regressor $P^\perp$) into two statistically independent components.
4. Identify the component that is the most normal.
5. Regress Y on P , X , and the identified normal component from ICA.

```
1 # Step 1: Set simulation parameters
2 N <- 1000                # number of observations
3 rho <- 0.8               # strength of endogeneity
4 alpha <- 2               # intercept
5 beta1 <- 1              # slope coefficient of endogenous regressor
6 beta2 <- 1              # slope coefficient of exogenous regressor
7 tau <- 1                 # effect of exogenous on endogenous regressor
8 # Step 2: Create data
9 # Exogenous regressor
10 X <- rnorm(N)
11 # Random noise (e.g., omitted factors)
12 xi <- rnorm(N)
13 # Exogenous influence (could be thought of as an instrument)
14 eta <- rgamma(N, shape = 1)
15 # Endogenous variable: combines error, exogenous component, and
    exogenous observed regressor
```

```

16 P <- rho * xi + eta + tau * X
17 # Outcome variable: depends on P, X, and the error
18 Y <- alpha + beta1 * P + beta2 * X + xi
19 # Step 4: Remove dependence between (Y, P) and X
20 mod1 <- lm(Y ~ X)
21 mod2 <- lm(P ~ X)
22 # Step 4: Apply Independent Component Analysis (ICA) on residuals
    from first-stage regression
23 data_matrix <- cbind(mod1$residuals, mod2$residuals)
24 ica_result <- ica(X = data_matrix, nc = 2, method = "jade")
25 # Step 4: Find the ICA component that looks most normally
    distributed
26 hist(ica_result$S[, 1])
27 hist(ica_result$S[, 2])
28 # Step 5: Estimate the regression model, controlling for the normal
    component
29 model <- lm(Y ~ P + X + ica_result$S[, 1])
30 # View the results
31 summary(model)

```

Listing 2: Simulation and estimation using ICA to address endogeneity in case of exogenous regressors.

3.3 Multiple endogenous and exogenous regressors

For multiple endogenous and exogenous regressors, the generalisation of the proposed approach is straightforward.

1. For each endogenous regressor $P_1, P_2, \dots, P_K$, regress it on all exogenous regressors $X_1, X_2, \dots, X_L$, and obtain the residuals $P_1^\perp, P_2^\perp, \dots, P_K^\perp$.

2. Regress dependent variable Y on all exogenous regressors $X_1, X_2, \dots, X_L$, and obtain the residuals $Y^\perp$.
3. Use ICA to decompose these residuals ($Y^\perp$ and $P_1^\perp, P_2^\perp, \dots, P_K^\perp$) into $(K + 1)$ statistically independent components.
4. Identify the component that is the most normal.
5. Regress Y on $P_1, P_2, \dots, P_K, X_1, X_2, \dots, X_L$, and the identified normal component from ICA.

4 Monte Carlo simulations

In the following, we conduct a set of Monte Carlo simulations to assess the finite sample performance of the proposed approach in various realistic scenarios. Furthermore, we benchmark the approach against rival methods like IV-based identification and IV-free approaches (such as the Gaussian copula estimator). We also conduct some robustness checks to assess if (and to what extent) the proposed approach is robust to deviations of (crucial) model assumptions.

4.1 Benchmark simulations

Consider the following DGP:

$$y_i = \alpha + \beta p_i + \gamma x_i + \xi_i, \text{ for } i = 1, \dots, n, \quad (21)$$

where p_i and x_i are endogenous and exogenous regressors, respectively, and ξ_i is the error term. To introduce endogeneity and to enable comparison with IV-based identification, we

generate

$$\xi_i \sim \mathcal{N}(0, 1), \quad v_i \sim \text{Ga}(1, 1) \quad (22)$$

$$x_i \sim \mathcal{N}(0, 1), \quad (23)$$

$$w_i \sim \text{Exp}(1), \quad (24)$$

$$p_i = v_i + \rho \xi_i + .5x_i + .5w_i. \quad (25)$$

In (25), the endogenous regressor is directly and linearly related to both the error and the exogenous regressor, i.e., no nonlinear transformation function is involved to introduce endogeneity. Furthermore, w_i serves as an instrument for implementing the IV estimator, and the degree of endogeneity is governed by ρ . We assume that the DGP is unknown and observe only y_i , p_i , x_i , and w_i (w_i is only used for IV estimation). We consider scenarios of exogeneity, various degrees of endogeneity, and different sample sizes, i.e., $\rho = \{0, .25, .5, 1\}$ and $n = \{250, 500, 1.000\}$.

Table 1 shows simulation results of least squares estimation without an endogeneity correction (OLS), the Park and Gupta (2012) Gaussian copula estimator, IV-based identification, and the proposed approach. OLS performs well when there is no endogeneity ($\rho = 0$), producing unbiased and rather accurate estimates. However, as endogeneity is introduced and increases, OLS estimates become increasingly biased. This bias is most evident in the estimates for the slope attached to the endogenous regressor, where values drift away from the true value, and for the intercept, which becomes increasingly underestimated. The root mean squared error (RMSE) increases with both ρ and N , suggesting that increasing the sample size does not help reduce bias. Furthermore, the corresponding t -statistics reveal systematic distortions in estimation that are not mitigated by larger sample sizes. These results confirm the well-known vulnerability of OLS to endogeneity.

In contrast, the IV estimator remains robust across all levels of endogeneity, and estimates for all parameters remain close to their true values. While there is a mild increase

N	$\rho = 0$					$\rho = .25$				$\rho = .5$				$\rho = 1$				
		Mean	SD	RMSE	t_{bias}	Mean	SD	RMSE	t_{bias}	Mean	SD	RMSE	t_{bias}	Mean	SD	RMSE	t_{bias}	
250	OLS	$\hat{\alpha}$	2.00	.105	.105	0.01	1.78	.109	.241	-1.99	1.60	.105	.411	-3.78	1.41	.103	.599	-5.79
		$\hat{\beta}$	1.00	.044	.044	-.002	1.12	0.05	.129	2.40	1.22	.055	.226	3.98	1.33	.058	.331	5.65
		$\hat{\gamma}$	.999	.067	.067	-.019	.938	.066	0.09	-0.94	0.89	.062	.127	-1.78	.841	0.06	.017	-2.63
	IV	$\hat{\alpha}$	2.00	.135	.135	.018	2.00	.137	.137	.019	2.00	.131	.131	.003	2.02	.142	.143	117
		$\hat{\beta}$	.999	.064	.064	-.012	.998	.066	.066	-.025	1.00	0.64	.064	0.00	.991	.071	.072	-.131
		$\hat{\gamma}$	.999	.071	.071	-.015	.999	.071	.071	-0.02	1.00	.069	.069	-.003	1.01	.074	.075	.156
	PG	$\hat{\alpha}$	2.01	0.26	0.26	0.02	2.41	.315	.521	1.32	2.70	.377	.793	1.85	2.81	.453	.928	1.79
		$\hat{\beta}$	.998	0.14	0.14	-.017	0.77	.171	.287	-1.34	.615	.208	.437	-1.85	.554	.248	0.51	-1.80
		$\hat{\gamma}$	.999	.093	.093	-.005	1.11	.109	.155	1.00	1.19	.129	.228	1.46	1.23	.156	.277	1.46
	ICA	$\hat{\alpha}$	2.01	.194	.194	.042	2.00	.187	.186	-.012	2.01	.194	.194	.059	2.04	.633	.634	.068
		$\hat{\beta}$	.996	.099	.099	-0.04	1.00	.096	.096	0.01	.994	.101	.101	-.062	.975	.379	.038	-.065
		$\hat{\gamma}$	1.00	0.08	.079	.007	.997	.078	.078	-.041	1.00	.078	.078	.037	1.02	.197	.198	.098
500	OLS	$\hat{\alpha}$	2.00	.072	.072	.002	1.79	.077	.222	-2.69	1.62	0.08	.391	-4.79	1.42	.077	0.591	-7.57
		$\hat{\beta}$	1.00	.032	.032	.018	1.12	.036	.121	3.23	1.21	.041	.217	5.13	1.32	.046	.326	7.04
		$\hat{\gamma}$	.998	.049	.049	-.045	.943	.047	.074	-1.19	.892	.045	.117	-2.39	.838	.042	.168	-3.86
	IV	$\hat{\alpha}$	2.00	.094	.094	.032	2.00	.093	.093	-.043	2.00	.091	.091	.036	2.01	.094	.094	.077
		$\hat{\beta}$	.999	.045	.045	-.023	1.00	.044	.044	.074	1.00	.044	.044	.005	.997	.045	.045	-.067
		$\hat{\gamma}$	.999	.051	.051	-.028	.999	.051	.051	-.014	1.00	.048	.048	-.047	1.00	.048	.048	0.03
	PG	$\hat{\alpha}$	2.00	.169	.169	.001	2.40	.224	.455	1.77	2.65	.286	.712	2.28	2.82	.374	.901	2.20
		$\hat{\beta}$	1.00	.091	.091	.008	.784	.121	.248	-1.79	.643	.158	0.39	-2.26	0.55	.204	.495	-2.20
		$\hat{\gamma}$	.998	.063	.063	-.037	1.11	.077	.132	1.39	1.17	.094	.197	1.84	1.22	.118	.253	1.89
	ICA	$\hat{\alpha}$	2.01	.133	.133	0.05	2.00	.127	.127	.015	2.00	.013	.013	.022	2.01	.137	.137	.069
		$\hat{\beta}$	.997	.069	.069	-.043	1.00	.066	.066	0.00	1.00	.066	.066	.007	.996	.071	.071	-.059
		$\hat{\gamma}$	.999	.058	.058	-0.01	1.00	.056	.056	.016	.998	.055	.055	-.042	1.00	.054	.054	.034
1000	OLS	$\hat{\alpha}$	2.00	0.05	0.05	-.042	1.79	.054	.218	-3.93	1.61	.061	.391	-6.28	1.42	.061	.585	-9.55
		$\hat{\beta}$	1.00	.021	.022	.045	1.12	.026	.118	4.44	1.21	.032	.215	6.54	1.32	.035	.321	9.01
		$\hat{\gamma}$	1.00	.033	.033	-.004	.942	.036	.068	-1.63	.895	.033	0.11	-3.18	.841	.032	.162	-4.96
	IV	$\hat{\alpha}$	2.00	.062	.062	-.056	2.00	.064	.065	-.064	2.00	.067	.067	.048	2.00	.065	.065	.048
		$\hat{\beta}$	1.00	0.03	0.03	.056	1.00	.032	.032	.047	.998	.032	.032	-.058	.998	.031	.031	-.051
		$\hat{\gamma}$	1.00	.034	.034	-.013	1.00	.037	.037	-0.02	1.00	.036	.036	0.05	1.00	.036	.036	.069
	PG	$\hat{\alpha}$	1.99	.109	.109	-.083	2.38	.145	.403	2.59	2.63	.224	.665	2.79	2.80	0.29	.853	2.77
		$\hat{\beta}$	1.00	.057	.058	.084	.792	.079	.222	-2.63	.655	.123	.366	-2.80	.559	.159	.468	-2.78
		$\hat{\gamma}$	.998	.043	.043	.047	1.10	.054	.117	1.90	1.17	.075	.188	2.28	1.22	0.91	0.24	2.43
	ICA	$\hat{\alpha}$	1.99	.092	.092	-.076	2.00	.088	.088	-.022	2.01	.095	.095	.054	2.00	.093	0.93	.047
		$\hat{\beta}$	1.00	.048	.048	.076	1.00	.046	.046	.007	.997	.049	.049	-0.06	.998	.049	.049	-.048
		$\hat{\gamma}$	.998	.039	.039	-.039	1.00	.041	.041	-.003	1.00	.041	.041	.059	1.00	0.04	0.04	0.07

Table 1: True values are $\alpha = 2$, and $\beta = \gamma = 1$. The table shows mean estimates (Mean), their empirical standard deviation (SD), root mean squared error (RMSE), and the t -ratio of the bias over $R = 1,000$ replications.

in standard errors as endogeneity rises, the IV estimator consistently delivers accurate inference, validating its suitability in the presence of endogenous regressors. The PG estimator performs similarly to OLS under exogeneity but fails to provide reliable estimates as endo-

geneity increases. Notably, the estimate of the slope is biased downward, while the intercept is overestimated. These biases grow with stronger endogeneity and are only partially mitigated by larger sample sizes. RMSE and bias statistics indicate that the PG estimator does not adequately correct for endogeneity in this setting (for similar findings, see Web Appendix A.11 in Haschka, 2022). This is to be expected, as the copula estimator requires a nonlinear regressor-error relationship, which is often difficult to justify.

The proposed estimator closely matches the performance of the IV estimator. Across all settings, it yields estimates with low bias and RMSE. While its standard errors are slightly larger than those of IV in smaller samples, they decrease with sample size. This suggests that ICA provides a viable and consistent alternative to IV-based identification, and also in scenarios where the Gaussian copula assumptions of a nonlinear regressor-error relation cannot be justified.

4.2 Multiple endogenous regressors

To show that the approach can be easily applied to scenarios with multiple endogenous and exogenous regressors, consider the following DGP:

$$y_i = \alpha + \beta_1 p_{1i} + \beta_2 p_{2i} + \beta_3 p_{3i} + \beta_4 p_{4i} + \gamma_1 x_{1i} + \gamma_2 x_{2i} + \xi_i, \quad (26)$$

$$\xi_i = \xi_{1i} + \xi_{2i}, \quad \xi_{1i} \sim \mathcal{N}(0, 1), \xi_{2i} \sim \mathcal{N}(0, 1), \quad (27)$$

$$x_{1i} \sim \text{Exp}(1), x_{2i} \sim \text{Exp}(1), \quad (28)$$

$$p_{1i} = v_{1i} + .5x_{1i} + \xi_{1i}, \quad (29)$$

$$p_{2i} = v_{2i} + \xi_{2i}, \quad (30)$$

$$p_{3i} = v_{3i} + \xi_{1i} + \xi_{2i}, \quad (31)$$

$$p_{4i} = v_{4i} + .5x_{1i} + .5x_{2i} + \xi_{1i} + \xi_{2i}, \quad (32)$$

$$v_{1i} \sim \text{Ga}(1, 1), v_{2i} \sim \text{Ga}(1, 1), v_{3i} \sim \text{Ga}(1, 1), v_{4i} \sim \text{Ga}(1, 1), \quad (33)$$

where the error term is composed of two normal components, one being related to p_{1i} and one to p_{2i} , respectively, and both being related to p_{3i} and p_{4i} . The exogenous regressor x_{1i} affects p_{1i} , and both x_{1i} and x_{2i} affect p_{4i} . The DGP is extensive because there are several endogenous variables correlated with single and/or multiple exogenous variables, and correlated with single or multiple components of the error.

		$N = 500$				$N = 1.000$			
		Mean	SD	RMSE	t_{bias}	Mean	SD	RMSE	t_{bias}
OLS	$\hat{\alpha}$	1.00	.035	.998	-28.5	1.00	.025	1.00	-40.41
	$\hat{\beta}_1$	1.17	0.03	.168	6.42	1.17	0.02	.168	8.93
	$\hat{\beta}_2$	1.17	.026	.168	6.36	1.167	.018	.168	9.22
	$\hat{\beta}_3$	1.34	.034	.337	9.81	1.34	.025	.335	13.2
	$\hat{\beta}_4$	1.33	.033	.336	10.1	1.33	.025	.334	13.43
	$\hat{\gamma}_1$	.749	.031	.253	-7.97	0.75	.023	.251	-10.95
	$\hat{\gamma}_2$	.833	.031	.169	-5.40	.834	.022	.168	-7.59
PG	$\hat{\alpha}$	2.63	.401	0.75	1.58	2.85	.309	0.90	2.73
	$\hat{\beta}_1$	.825	.158	.236	-1.10	.809	.113	.222	-1.69
	$\hat{\beta}_2$	.787	.145	.257	-1.46	.789	.109	2.38	-1.93
	$\hat{\beta}_3$	.782	.254	.335	-8.58	.703	.191	.353	-1.56
	$\hat{\beta}_4$	.978	.266	.267	1.26	.857	.192	.239	1.73
	$\hat{\gamma}_1$	1.10	0.16	.187	.609	1.16	.129	.209	1.28
	$\hat{\gamma}_2$	1.00	.157	.157	.056	1.07	.115	.134	.604
Proposed	$\hat{\alpha}$	2.02	1.98	1.98	.013	2.03	.241	.242	.112
	$\hat{\beta}_1$	.999	.566	.566	-.002	.997	0.13	0.13	-.027
	$\hat{\beta}_2$	1.02	1.50	1.50	.014	1.00	0.14	.142	-.001
	$\hat{\beta}_3$	.982	.266	.266	-.068	.992	.107	.107	-.078
	$\hat{\beta}_4$	.991	.468	.468	-0.02	.987	0.11	.111	-.115
	$\hat{\gamma}_1$	1.00	.407	.407	.019	1.00	.096	.069	.079
	$\hat{\gamma}_2$	1.00	.356	.256	.016	1.00	.071	.071	.087

Table 2: True values are $\alpha = 2$, and all $\beta =$ all $\gamma = 1$. For further notes, see Table 1.

Table 2 evaluates the performance of three estimators considered. OLS performs poorly and clearly fails to address the endogeneity in this multivariate setting. The PG estimator improves upon OLS but still fails to eliminate bias. It tends to underestimate the coefficients on the endogenous variables, with inconsistent patterns depending on the regressor.

Although the intercept is much closer to the true value compared to OLS, the estimates remain imprecise and display notably high RMSE. For the exogenous regressors, the estimates are closer to the true values, but are still significantly biased.

By contrast, the proposed ICA estimator performs well even in this high-dimensional setting. It produces estimates with minimal bias across all coefficients, including the intercept, the four endogenous variables, and the two exogenous regressors. The RMSE is low and decreases with sample size, and the t -ratios values are consistently close to zero, indicating negligible systematic distortion. The precision of the estimates increases substantially from $N = 500$ to $N = 1,000$, as evident through reduced standard deviations. ICA accurately recovers all parameters, demonstrating its scalability to multivariate cases with multiple sources of endogeneity.

4.3 A true omitted variables setting

To show that the proposed approach can handle the traditional omitted variables bias, consider the following DGP:

$$y_i = \alpha + \beta p_i + \sum_{j=1}^J \gamma_j \omega_{ij} + \xi_i, \quad \xi_i \sim \mathcal{N}(0, 1) \quad (34)$$

$$p_i = v_i + \delta \sum_{j=1}^J \gamma_j \omega_{ij}, \quad v_i \sim \text{Ga}(1, 1), \quad (35)$$

where the $J = \{20, 10, 5\}$ ω_{ij} variables are generated independently (Scenario 1) using uniform, exponential, binomial, Poisson, Student-t, χ^2 , Beta, Gamma, lognormal, and Weibull distributions. These variables are used in the generation of y and p , but do not enter the estimation. Hence, the regression error becomes $\xi_i^* = \sum_{j=1}^J \gamma_j \omega_{ij} + \xi_i$, where ξ_i is exogenous. This setting is particularly interesting, because none of the J ω_{ij} variables is normally distributed. According to the CTL, their sum should sum up to a normal distribution. Thus, we regress y_i on p_i and omit the J ω_{ij} variables in the estimation, rendering z_i endogenous.

We set $\alpha = 2$, $\beta = 1$, and $\delta = 1$. The correlation between p and the variables ω_j ranges from .15 to .25 on average. Furthermore, we consider a second scenario (Scenario 2) with omitted variables ω_{ij} being correlated (to demonstrate that the proposed approach still works because the CTL applies also for correlated random variables). For this purpose, we first generate the J ω_{ij} variables which are then subjected to perturbation such that they are all correlated with each other by .5 (note that this does not change their marginal distribution, see Section 4.6.2 for details).

			Independent OV				Dependent OV			
J			Mean	SD	RMSE	t_{bias}	Mean	SD	RMSE	t_{bias}
20	OLS	$\hat{\alpha}$	2.00	.033	.033	.026	2.00	.052	.052	-.039
		$\hat{\beta}$	2.68	.164	1.68	-10.2	7.77	.210	6.78	-32.3
	PG	$\hat{\alpha}$	-3.58	.657	5.62	8.50	-5.75	3.14	8.36	2.47
		$\hat{\beta}$	-.003	.359	2.04	2.79	5.34	1.10	3.52	-3.95
	Proposed	$\hat{\alpha}$	2.00	.032	.032	.026	2.00	.138	.138	-.027
		$\hat{\beta}$	.991	.321	1.06	.029	.943	1.03	1.48	.055
10	OLS	$\hat{\alpha}$	2.00	.032	.032	.029	2.00	.069	.069	.001
		$\hat{\beta}$	1.91	.116	.919	-7.86	4.55	.216	3.56	-16.4
	PG	$\hat{\alpha}$	-1.36	.424	3.39	7.92	-7.29	1.30	9.39	7.12
		$\hat{\beta}$	.252	.247	1.77	3.03	.522	.661	1.62	.723
	Proposed	$\hat{\alpha}$	2.00	.032	.032	.030	2.00	.101	.101	-.005
		$\hat{\beta}$	.991	.240	1.03	.036	.997	.505	1.12	.004
5	OLS	$\hat{\alpha}$	2.00	.031	.031	-.015	2.00	.056	.056	.022
		$\hat{\beta}$	1.84	.082	.843	-10.2	2.88	.115	1.88	-16.4
	PG	$\hat{\alpha}$	-.760	.353	2.78	7.83	-3.55	.636	5.59	8.72
		$\hat{\beta}$	.517	.193	1.49	2.49	.497	.309	1.53	1.63
	Proposed	$\hat{\alpha}$	2.00	.032	.032	-.017	1.99	.081	.081	.013
		$\hat{\beta}$	.995	.178	1.02	.026	1.00	.283	1.04	-.007

Table 3: True values are $\alpha = 2$, and $\beta = 1$. For further notes, see Table 1.

Table 3 examines how well the proposed ICA estimator handles omitted variable (OV)

bias, in case of which the omitted variables are mutually independent (Scenario 1), and in which they are correlated (Scenario 2). Note that in both cases, the dependent variable y_i is influenced by the latent set of variables ω_{ij} that also affect the regressor p_i , but these variables are not included in the estimation. As a result, the regression error term effectively absorbs them, leading to endogeneity. The number of omitted variables, J , varies across simulations: 20, 10, and 5. Importantly, these ω_{ij} variables are drawn from a wide range of non-Gaussian distributions, and yet, under the Central Limit Theorem (CLT), their sum behaves approximately normally.

The OLS estimator consistently overestimates β across all values of J , with the bias becoming more severe as the number of omitted variables increases. In the independent OV scenario with $J = 20$, the estimated β is almost triple of the true value. In the dependent OV scenario, the distortion is even more extreme. The standard errors are low, which misleadingly suggests precision, but the RMSE and t -ratio values reveal massive bias. These findings underscore the vulnerability of OLS to omitted variable bias, especially when many unobserved factors drive both the regressor and the outcome.

The PG estimator offers no real remedy to this problem. In fact, its performance is erratic and unstable. The estimates for β are either near zero or even negative in the independent case, and wildly inaccurate in the dependent scenario. For instance, with $J = 20$ the PG estimator delivers a estimate close to zero in Scenario 1 and a wildly inflated value of 5.34 in Scenario 2. The intercept estimates are also implausible. These results suggest that the PG estimator is not suitable for handling omitted variable bias in the traditional understanding.

By contrast, the proposed ICA estimator performs robustly and consistently across all scenarios and values of J . It delivers nearly unbiased estimates for both the intercept and the slope, regardless of whether the omitted variables are independent or correlated. For instance, in the most challenging case (dependent OVs with $J = 20$), the ICA estimate for β is .943 with an RMSE of just 1.48, a stark contrast to the extreme overestimation by OLS and the instability of PG. As the number of omitted variables decreases (from 20 to 5), the

ICA estimator maintains its accuracy, and does not exhibit bias.

These results demonstrate that the ICA estimator successfully corrects for endogeneity caused by omitted variables in the traditional understanding, even when those variables are non-Gaussian and mutually dependent. The approach benefits from the CTL, which ensures the approximately Gaussian nature of the aggregated error, enabling ICA to isolate the endogenous component effectively. In doing so, the method offers a powerful solution a powerful solution to endogeneity because of omitted variables.

4.4 Comparison with other IV-free approaches

Next, we compare the performance of the proposed approach with three alternative estimators. However, it should be noted that this is not a fair comparison in any case, as the assumptions about the regressor-error relationship differ fundamentally.

4.4.1 Comparison with copula-based endogeneity corrections

Comparison with PG (Park and Gupta, 2012) Consider the following DGP:

$$y_i = \alpha + \beta p_i + \xi_i, \quad (36)$$

$$p_i = \exp\{v_i + \rho\Phi(\xi_i)\}, \quad (37)$$

$$\xi_i \sim \mathcal{N}(0, 1), \quad v_i \sim \mathcal{N}(0, 1). \quad (38)$$

Following the identification requirements of copula approaches, we use a nonlinear transformation function to relate the normal error to the endogenous regressor. Since the proposed approach requires exogenous variation on the endogenous regressor, we put a noise component in the exponential function. Since this is additive *within* the nonlinear transformation function, identification requirements by the copula estimator are not violated (Qian et al., 2024). While the copula model is fully identified, the identifying assumptions of the proposed approach are violated because the endogenous regressor is not additively decomposable. Al-

though it is therefore not a fair comparison, we continue the example.

Table 4 compares the proposed ICA estimator with the copula-based estimator by Park and Gupta (PG). Although the identifying assumptions of the proposed ICA approach are violated, it performs well. When endogeneity is absent ($\rho = 0$), all estimators produce unbiased estimates and RMSEs are small. Once endogeneity is introduced ($\rho = 1$), OLS becomes biased, but PG recovers the slope coefficient well and rather accurately. The ICA estimator, while not perfectly aligned with the DGP's structure, still delivers accurate slope estimates with minimal bias and low RMSE. The reason why the proposed approach still works is the following. Recall that the proposed approach tries to decompose the endogenous regressor into a normal and a nonnormal component, where the normal component is assumed to be the endogenous part. In the DGP, $\exp\{\Phi(\xi_i)\}$ is more close to normality than $\exp\{v_i\}$, such that the correct endogenous part is identified. Overall, even in a setting designed for the copula model and against ICA's assumptions, the proposed method proves surprisingly robust and delivers competitive performance.

			$N = 500$				$N = 1.000$			
			Mean	SD	RMSE	t_{bias}	Mean	SD	RMSE	t_{bias}
$\rho = 0$	OLS	$\hat{\alpha}$	2.00	.162	.162	-.004	2.01	.116	.116	.053
		$\hat{\beta}$	1.00	.091	0.09	.006	.996	.065	.065	-.059
	PG	$\hat{\alpha}$	2.02	0.65	0.65	.026	1.99	0.44	0.44	-.013
		$\hat{\beta}$	.979	.732	.732	-.028	1.01	.469	.495	.022
	Proposed	$\hat{\alpha}$	2.00	.235	.235	0.00	2.01	.158	.159	.057
		$\hat{\beta}$	1.00	.136	.136	.001	.995	0.09	0.09	-.061
$\rho = 1$	OLS	$\hat{\alpha}$	1.45	.105	.564	-5.29	1.46	.083	.551	-6.57
		$\hat{\beta}$	1.2	0.04	.201	4.95	1.19	.031	.196	6.25
	PG	$\hat{\alpha}$	.448	0.07	1.55	-22.3	0.45	.051	1.55	-30.4
		$\hat{\beta}$	1.04	.008	.045	5.69	1.04	.006	.044	7.27
	Proposed	$\hat{\alpha}$	1.73	.085	.281	-3.17	1.76	.071	.249	-3.36
		$\hat{\beta}$	1.10	.028	.099	3.42	1.08	.024	.088	3.59

Table 4: True values are $\alpha = 2$, and $\beta = 1$. For further notes, see Table 1

Comparison with 2sCOPE (Yang et al., 2024) Consider the following DGP:

$$y_i = \alpha + \beta p_i + \gamma x_i + \xi_i, \quad (39)$$

$$p_i = \exp\{v_i + \Phi(p_i^*)\}, \quad v_i \sim \mathcal{N}(0, 1) \quad (40)$$

$$\begin{pmatrix} p_i^* \\ \xi_i \\ x_i \end{pmatrix} \sim \mathcal{N} \left(\begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 & \rho & .3 \\ \rho & 1 & 0 \\ .3 & 0 & 1 \end{bmatrix} \right). \quad (41)$$

In this scenario, a multivariate normal distribution is used to introduce dependence among the components. According to the requirements of the copula approach, a nonlinear transformation function is used to generate regressor-error dependence. While the copula model is again fully identified, the proposed approach lacks identification for the same reason as in the scenario before.

			$N = 500$				$N = 1.000$			
			Mean	SD	RMSE	t_{bias}	Mean	SD	RMSE	t_{bias}
$\rho = 0$	OLS	$\hat{\alpha}$	2.00	.056	.056	-.005	2.00	0.04	0.04	.024
		$\hat{\beta}$	1.00	.012	.012	-.005	1.00	.008	.008	-.055
	2sCOPE	$\hat{\alpha}$	2.00	.073	.073	-0.01	2.00	0.05	0.05	0.01
		$\hat{\beta}$	1.00	.021	.021	.011	1.00	.013	.013	-.026
	Proposed	$\hat{\alpha}$	2.00	.071	.071	-0.24	2.00	.051	.051	0.01
		$\hat{\beta}$	1.00	0.02	0.02	.024	1.00	.014	.014	-0.02
$\rho = .8$	OLS	$\hat{\alpha}$	1.88	0.06	.138	-2.08	1.88	.044	.127	-2.69
		$\hat{\beta}$	1.05	.015	.047	3.04	1.04	.011	.044	3.79
	2sCOPE	$\hat{\alpha}$	2.00	0.07	0.07	0.02	2.00	.046	.046	0.08
		$\hat{\beta}$	1.00	.019	.019	-.017	.999	.012	.012	-.086
	Proposed	$\hat{\alpha}$	1.95	0.07	.089	-7.85	1.95	.049	.069	-.973
		$\hat{\beta}$	1.02	.019	.027	1.05	1.01	.013	.022	1.28

Table 5: True values are $\alpha = 2$, and $\beta = 1$. For further notes, see Table 1.

Table 1 evaluates the proposed method against 2sCOPE. The nonlinear transformation

introduces endogeneity, aligning with the assumptions of 2sCOPE but not of the proposed approach, which requires an additive structure. When there is no endogeneity ($\rho = 0$), all estimators produce unbiased results with minimal error. Once endogeneity is introduced ($\rho = .8$), both 2sCOPE and the proposed method can correct the bias. While 2sCOPE performs slightly better overall, the proposed approach still recovers the slope accurately and approximates the intercept well despite lacking formal identification.

4.4.2 Comparison with the latent IV approach by Ebbes et al. (2009)

Finally, we compare the proposed approach to the method of latent instrumental variables (LIV). This method is the most similar extant instrument-free approach, since it assumes that the endogenous regressor can be decomposed additively into an endogenous normal part (just like in our approach), but the exogenous part has to be binary or discrete. Consider the following DGP:

$$y_i = \alpha + \beta p_i + \gamma x_i + \xi_i, \quad (42)$$

$$p_i = v_i + e_i, \quad (43)$$

$$\begin{pmatrix} \xi_i \\ e_i \\ m_i \end{pmatrix} \sim \mathcal{N} \left(\begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 & .5 & 0 \\ .5 & 1 & r \\ r & 0 & 1 \end{bmatrix} \right), \quad (44)$$

$$v_i = \begin{cases} 0 & \text{if } m_i < 0, \\ 4 & \text{if } m_i \geq 0. \end{cases} \quad (45)$$

Note that the LIV method is based on two key assumptions: v_i is a discrete variable and v_i is exogenous. The second assumption is similar to the exclusion restriction in usual IV models and in the proposed approach (note that v_i is unobserved). In the DGP, we consider $r = \{0, .5\}$, which implies that the exogenous component in the endogenous regressor is not *strictly* exogenous, but only exogenous *conditional* on the endogenous component. The aim

is to show that in the proposed approach, the exogeneity assumption for v_i can be relaxed from strict to conditional.

			$N = 500$				$N = 1,000$			
			Mean	SD	RMSE	t_{bias}	Mean	SD	RMSE	t_{bias}
$r = 0$	OLS	$\hat{\alpha}$	1.80	.059	0.21	-3.41	1.80	.041	.205	-4.91
		$\hat{\beta}$	1.10	.019	.102	5.36	1.10	.013	0.10	7.62
	LIV	$\hat{\alpha}$	2.02	.104	.106	.023	2.02	.083	.084	0.18
		$\hat{\beta}$	.994	.035	.035	-.182	.996	.028	.028	-.142
	Proposed	$\hat{\alpha}$	2.00	.071	.071	-.048	2.00	.048	.047	-0.02
		$\hat{\beta}$	1.00	.025	.025	.036	.999	.018	.018	-.032
$r = .5$	OLS	$\hat{\alpha}$	1.85	.054	.161	-2.80	1.85	.039	.155	-3.83
		$\hat{\beta}$	1.08	.016	.077	4.64	1.07	.012	.077	6.37
	LIV	$\hat{\alpha}$	2.05	.121	0.13	0.41	2.05	.121	.131	0.42
		$\hat{\beta}$	.998	.036	.038	-.335	.987	.036	.038	-.362
	Proposed	$\hat{\alpha}$	2.00	.058	.058	-.016	2.00	.042	.042	.024
		$\hat{\beta}$	1.00	.019	.019	-.016	1.00	.014	.014	-.003

Table 6: True values are $\alpha = 2$, and $\beta = 1$. For further notes, see Table 1.

Table 6 contrasts the proposed estimator with the LIV method. LIV assumes an additive decomposition of the endogenous regressor into a normal endogenous component and a discrete exogenous one. It also requires strict exogeneity of the latter, which resembles the exclusion restriction in IV frameworks. In the data-generating process, this condition holds either strictly ($r = 0$) or only conditionally ($r = .5$). LIV performs well when strict exogeneity holds but becomes slightly biased when exogeneity is only conditional (for similar findings, see Park and Gupta, 2012). In contrast, the proposed method remains unbiased in both settings, and delivers small RMSE across sample sizes. These results suggest that the proposed estimator does not require strict exogeneity of the exogenous component, only conditional independence, thus generalizing LIV’s identification condition.

4.5 Adverse effects of breakdown in model identification

We next quantify the adverse effects caused by a breakdown in model identification. The crucial identifying assumptions are (i) that the endogenous regressor's exogenous component is non-normal, and (ii) that this component is conditionally exogenous, i.e., uncorrelated with the error term given the endogenous regressor.

4.5.1 Exogenous component becomes normal

Consider the following DGP:

$$y_i = \alpha + \beta p_i + \xi_i, \quad (46)$$

$$p_i = v_i + \rho \xi_i, \quad (47)$$

$$\text{Scenario 1: } v_i \sim (\chi^2(k) - k) / \sqrt{2k}, \quad (48)$$

$$\text{Scenario 2: } v_i \sim t(df) / \sqrt{(df / (df - 2))}. \quad (49)$$

We consider $k = \{15, 30, 60, 120\}$, and $df = \{4, 8, 16, 32\}$, such that v_i approaches a normal distribution in a 'stepwise' manner. In both scenarios, v_i is scaled to have zero mean and unit variance such that its first two moments are identical to those of ξ_i . Moreover, we consider scenarios of exogeneity, moderate, and strong endogeneity, i.e., $\rho = \{0, .5, 1\}$.

				$\rho = 0$				$\rho = .5$				$\rho = 1$			
				Mean	SD	RMSE	t_{bias}	Mean	SD	RMSE	t_{bias}	Mean	SD	RMSE	t_{bias}
Scenario 1	$k = 15$	OLS	$\hat{\alpha}$	2.00	.031	.031	-.039	2.00	.029	.029	-0.01	2.00	.029	.029	-0.01
			$\hat{\beta}$	1.00	0.03	.031	-.004	1.40	.028	.402	14.4	1.40	.028	.402	14.4
		Proposed	$\hat{\alpha}$	2.00	.033	.033	-0.03	2.00	.034	.034	.031	2.00	.034	.034	.031
			$\hat{\beta}$	1.00	.258	.258	.024	.971	.284	.286	-.102	.971	.284	.286	-.102
	$k = 20$	OLS	$\hat{\alpha}$	2.00	.031	.031	.019	2.00	.029	.029	-.039	2.00	.022	.022	.038
			$\hat{\beta}$	1.00	.032	.032	.014	1.40	.026	.401	15.1	1.50	.017	.501	29.8
		Proposed	$\hat{\alpha}$	2.00	.034	.034	.015	2.00	.035	.035	-.059	1.99	.423	.423	-.034
			$\hat{\beta}$	1.00	.343	.343	-.014	.956	0.36	.363	-.121	1.03	8.10	8.09	.003
	$k = 25$	OLS	$\hat{\alpha}$	2.00	.038	.038	-.024	1.72	.032	.284	-8.71	1.65	.024	.335	-15.0
			$\hat{\beta}$	1.00	.031	.031	.051	1.40	.025	.401	15.9	1.50	.017	.501	30.1
		Proposed	$\hat{\alpha}$	2.00	.027	.027	-.008	1.78	8.95	8.95	-.025	2.11	.945	.951	.115
			$\hat{\beta}$	1.00	.379	.379	.006	1.28	11.5	11.4	.025	.845	1.30	1.30	-0.12
	$k = 30$	OLS	$\hat{\alpha}$	2.00	.032	.032	0.04	2.00	0.03	0.03	-.012	2.00	0.03	0.03	-.012
			$\hat{\beta}$	.999	0.03	.032	-0.21	1.40	.026	.401	15.4	1.40	.026	.401	15.36
		Proposed	$\hat{\alpha}$	2.00	0.04	0.04	.036	2.00	.138	.138	.003	2.00	.138	.138	.003
			$\hat{\beta}$	1.05	1.11	1.11	.046	.694	4.81	4.82	-.063	.694	4.81	4.82	-.063
Scenario 2	$df = 4$	OLS	$\hat{\alpha}$	2.00	.032	.032	.026	2.00	.026	.026	0.03	2.00	.026	.026	0.03
			$\hat{\beta}$	.1.00	.045	.045	-.018	1.67	.053	.673	12.6	1.67	.053	.673	12.6
		Proposed	$\hat{\alpha}$	2.00	.032	.032	.028	2.00	.032	.032	.059	2.00	.032	.032	.059
			$\hat{\beta}$	.998	.113	.113	-.021	.993	.118	.118	-0.06	.993	.118	.118	-.06
	$df = 8$	OLS	$\hat{\alpha}$	2.00	.031	.031	.059	1.99	.028	.028	-.026	2.00	.028	.028	-.026
			$\hat{\beta}$	1.00	.037	.037	-.005	1.50	.032	.501	15.63	1.50	.032	.501	15.63
		Proposed	$\hat{\alpha}$	2.01	0.35	0.35	.025	2.00	.035	.035	.007	2.00	.035	.035	.007
			$\hat{\beta}$	.888	13.6	13.6	-.008	1.07	.568	.573	.126	1.07	.568	.573	.126
	$df = 12$	OLS	$\hat{\alpha}$	2.00	.032	.032	.006	2.00	.028	.028	.046	2.00	.021	.021	-.007
			$\hat{\beta}$	1.00	.035	.035	.005	1.46	.028	.461	16.3	1.55	.016	.546	33.7
		Proposed	$\hat{\alpha}$	2.01	.282	.282	.031	2.00	.162	.162	-.024	2.00	0.05	0.05	-.015
			$\hat{\beta}$	.985	12.7	12.6	-.001	1.17	8.00	7.99	.021	1.16	1.00	1.01	.157
	$df = 16$	OLS	$\hat{\alpha}$	2.00	.031	.031	-.005	2.00	.028	.028	-.004	2.00	.028	.028	-.004
			$\hat{\beta}$	1.00	.034	.034	.025	1.44	.028	.445	16.07	1.44	.028	.445	16.07
		Proposed	$\hat{\alpha}$	2.03	1.22	1.22	.027	2.00	0.13	0.13	-.014	2.00	0.13	0.13	-.014
			$\hat{\beta}$	3.45	64.4	64.4	.038	1.82	11.92	11.94	.068	1.81	11.9	11.9	.068

Table 7: True values are $\alpha = 2$, and $\beta = 1$. For further notes, see Table 1.

Table 7 assesses how the proposed estimator responds when its key identifying assumption, the non-normality of the exogenous component in the endogenous regressor, is progressively violated. When v_i is strongly non-normal (e.g., $k = 15$ or $df = 4$), the proposed method performs well, yielding estimates with low bias and RMSE across all endogeneity levels. However, as v_i approaches normality (larger k or df), the model gradually loses its ability to isolate the endogenous component. This breakdown becomes apparent in the estimated slope coefficients, which show growing variability and in some cases substantial bias, particularly under

strong endogeneity. The results confirm that the estimator’s performance hinges critically on the non-normality of the exogenous component. As this assumption is relaxed, identification deteriorates because the approach can no longer separate the endogenous normal component for the exogenous normal component.

4.5.2 Exogenous component becomes correlated

Consider the following DGP:

$$y_i = \alpha + \beta p_i + \xi_i, \quad (50)$$

$$\begin{pmatrix} p_i^* \\ \xi_i \end{pmatrix} \sim \mathcal{N} \left(\begin{bmatrix} 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 & \rho \\ \rho & 1 \end{bmatrix} \right), \quad (51)$$

$$p_i = F_{\text{Ga}(1,1)}^{-1}(\Phi(p_i^*)) + \xi_i, \quad (52)$$

where $F_{\text{Ga}(1,1)}^{-1}$ is the inverse cdf of the Gamma(1,1) distribution. To introduce correlation between the nonnormal component in the endogenous regressor and the error, we consider $\rho = \{.1, .2, .4, .8\}$. In this setting, the non-normal component in the endogenous regressor becomes correlated with the error term.

		$\rho = .1$				$\rho = .2$				$\rho = .4$				$\rho = .8$			
		Mean	SD	RMSE	t_{bias}	Mean	SD	RMSE	t_{bias}	Mean	SD	RMSE	t_{bias}	Mean	SD	RMSE	t_{bias}
OLS	$\hat{\alpha}$	1.50	.023	.501	-21.3	1.50	.022	.501	-23.0	1.49	0.02	.501	-25.5	1.49	.013	.501	37.13
	$\hat{\beta}$	1.50	.023	.502	22.2	1.50	.021	.501	24.4	1.50	.018	0.50	27.9	1.50	.012	.501	42.2
Proposed	$\hat{\alpha}$	1.96	.073	.084	-.575	1.91	.064	.019	-1.38	1.84	.052	.173	-3.16	1.70	0.03	0.03	-10.0
	$\hat{\beta}$	1.04	.067	.079	.632	1.09	.057	.105	1.53	1.16	.047	.169	3.49	1.29	.024	.299	12.3

Table 8: True values are $\alpha = 2$, and $\beta = 1$. For further notes, see Table 1.

Estimation results are documented in Table 8. The proposed estimator can still correct for the endogeneity bias if there is small correlation between the non-normal and the endogenous component. The higher the correlation, the stronger the bias in the slope and the intercept. However, it is noticeable that in the case of high correlation, the proposed estimator is still less biased than the OLS estimator. This shows that even if the important assumption of

exogeneity of the non-normal component is violated, the proposed estimator performs better than OLS.

4.6 Robustness checks

4.6.1 Copula-type regressor-error dependence

The dependence between the endogenous regressor and the error is a consequence of an economic mechanism (such as marketing strategy decision). The proposed approach does not require any assumption about the regressor-error relationship - in contrast to copula-based approaches. To show that performance of the proposed approach remains unaffected by changes in the regressor-error dependence, in the following, we generate the dependence using several (archimedean) copula models. Consider the following DGP

$$y_i = \alpha + \beta p_i + \xi_i, \quad (53)$$

$$p_i = v_i + \Phi^{-1}(u_p), \quad v_i \sim \text{Ga}(1, 1) - 1, \quad (54)$$

$$\xi_i = \Phi^{-1}(u_\xi), \quad (55)$$

and we consider the following copula models to generate the pairs (u_p, u_ξ) :

(a) Ali-Mikhail-Haq copula (with $\theta = 1$):

$$C(u_{p,i}, u_{\xi,i}) = \frac{u_{p,i}u_{\xi,i}}{1 - \theta(1 - u_{p,i})(1 - u_{\xi,i})}.$$

(b) Plackett copula (with $\theta = 20$):

$$C(u_{p,i}, u_{\xi,i}) = \left(1 + (\theta - 1)(u_{p,i} + u_{\xi,i}) - \sqrt{(1 + (\theta - 1)(u_{p,i} + u_{\xi,i}))^2 - 4u_{p,i}u_{\xi,i}\theta(\theta - 1)} \right) (2(\theta - 1))^{-1}$$

(c) Farlie-Gumbel-Morgenstern copula (with $\theta = 1$):

$$C(u_{p,i}, u_{\xi,i}) = u_{p,i}u_{\xi,i}(1 + \theta(1 - u_{p,i})(1 - u_{\xi,i}))$$

(d) Clayton copula (with $\theta = 2$):

$$C(u_{p,i}, u_{\xi,i}) = (u_{p,i}^\theta + u_{\xi,i}^\theta - 1)^{-1/\theta}$$

(e) Frank copula (with $\theta = 2$):

$$C(u_{p,i}, u_{\xi,i}) = -\frac{1}{\theta} \log \left(1 + \frac{(\exp\{-\theta u_{p,i}\} - 1)(\exp\{-\theta u_{\xi,i}\} - 1)}{\exp\{-\theta\} - 1} \right)$$

(f) Cauchy copula ($\rho = .7$):

$$C(u_{p,i}, u_{\xi,i}) = \int_{-\infty}^{t^{-1}(u_{p,i})} \int_{-\infty}^{t^{-1}(u_{\xi,i})} \frac{1}{2\pi\sqrt{1-\rho^2}} \left(1 + \frac{x^2 - 2\rho xy + y^2}{(1-\rho^2)^2} \right)^{-3/2} dx dy, \text{ with } t^{-1} \text{ denoting the generalized inverse function of the Cauchy distribution.}$$

Because the dependence structure is fixed at the data generation stage, the location of the exogenous component shifts relative to the endogenous component. If this shift is not explicitly modeled (i.e., if the offset in v_i is not accounted for), the intercept estimate becomes systematically biased. This is why we centre v_i in (54). Importantly, this does not affect the slope estimate, which is the primary object of interest in causal inference.

		AMH				Placket				FGM			
		Mean	SD	RMSE	t_{bias}	Mean	SD	RMSE	t_{bias}	Mean	SD	RMSE	t_{bias}
OLS	$\hat{\alpha}$	2.00	.029	.029	0.03	2.00	.024	.024	.021	2.00	0.03	0.03	-.001
	$\hat{\beta}$	1.25	.025	.249	10.1	1.38	.024	.377	15.9	1.16	.023	.161	7.08
Proposed	$\hat{\alpha}$	2.00	.056	.056	.023	2.00	.059	.059	.039	2.00	0.05	0.05	-.011
	$\hat{\beta}$	.961	.111	.118	-.347	.945	.107	0.12	-.511	.995	.107	.107	-.048
		Clayton				Frank				Cauchy			
		Mean	SD	RMSE	t_{bias}	Mean	SD	RMSE	t_{bias}	Mean	SD	RMSE	t_{bias}
OLS	$\hat{\alpha}$	2.00	.026	.026	-.021	2.00	.031	.031	0.00	2.00	.025	.025	.024
	$\hat{\beta}$	1.34	.023	.344	14.6	1.15	.022	1.52	6.75	1.35	.024	.346	14.1
Proposed	$\hat{\alpha}$	2.00	.058	.058	-0.05	2.00	.052	.051	.025	2.00	.057	.057	-.034
	$\hat{\beta}$	.959	.105	.112	-.389	.974	.109	.112	-.241	.924	0.11	.133	-.689

Table 9: True values are $\alpha = 2$, and $\beta = 1$. For further notes, see Table 1.

As shown in Table 9, the proposed estimator recovers the true slope coefficient accurately across all copula types; the estimates are very close to the true value of 1, with relatively small RMSE and negligible statistical bias. The only exception arises in the case of the Cauchy copula, which imposes a particularly heavy-tailed and nonlinear dependence structure; here, the estimator is slightly biased, though it still performs substantially better than OLS.

4.6.2 Nonparametric regressor-error dependence

In copula-based approaches, Eckert and Hohberger (2022) find that the Gaussian copula fails to model dependence structure that are not generated by parametric dependence generators, leading to biased estimates. To further substantiate that the proposed approach is not affected by how the regressor-error dependence is introduced, we consider the following DGP:

$$y_i = \alpha + \beta p_i + \xi_i, \quad (56)$$

$$p_i = v_i + p_i^*, \quad v_i \sim \text{Ga}(1, 1) - 1. \quad (57)$$

We first draw $p_i^* \sim \mathcal{N}(0, 1)$ and $\xi_i \sim \mathcal{N}(0, 1)$, and then introduce nonparametric dependence using the algorithm by Ruscio and Kaczetow (2008). Specifically, the algorithm rotates the model component matrix (p_i^*, ξ_i) until empirical correlations are $\text{Corr}[\xi, p^*] = .8$ (measured either by Pearson’s or Spearman’s coefficient) while the marginal distributions are kept unchanged. Hence, data is simulated and correlations are introduced without assuming parametric dependence (for similar setup, see Eckert and Hohberger, 2022).

		Pearson				Spearman			
		Mean	SD	RMSE	t_{bias}	Mean	SD	RMSE	t_{bias}
OLS	$\hat{\alpha}$	1.60	.054	.404	-7.36	1.61	.052	.392	-7.47
	$\hat{\beta}$	1.40	.024	.401	16.4	1.38	.023	0.39	16.7
Proposed	$\hat{\alpha}$	3.00	0.07	1.00	14.2	3.00	0.07	1.00	14.4
	$\hat{\beta}$	.993	0.08	0.08	-.888	1.02	0.07	.073	.297

Table 10: True values are $\alpha = 2$, and $\beta = 1$. For further notes, see Table 1.

Table10 reports the results. The proposed estimator performs well under both correlation measures. It yields estimates that are close to the true parameters, with low bias and RMSE. These findings underscore the flexibility of the proposed method: it accommodates arbitrary dependence structures without relying on parametric assumptions, thereby avoiding

the pitfalls of Gaussian copula-based approaches. Note that the Gaussian copula approach is not able to handle nonparametric dependence, as demonstrated by (Eckert and Hohberger, 2022).

4.7 Varying the distribution of the regressor's exogenous component

Here, we test if the proposed approach is sensitive to the type of non-Gaussianity in the exogeneous component.

Consider the following DGP:

$$y_i = \alpha + \beta p_i + \xi_i, \quad \xi_i \sim \mathcal{N}(0, 1) \quad (58)$$

$$p_i = v_i + \xi_i, \quad (59)$$

$$\text{Scenario 1: } v_i \sim (\chi^2(3) - 3) / \sqrt{6}, \quad (60)$$

$$\text{Scenario 2: } v_i \sim (\text{Lognormal}(0, 1) - e^{1/2}) / \sqrt{e^2 - e^1} \quad (61)$$

$$\text{Scenario 3: } v_i \sim \text{Unif}[-1, 1] / \sqrt{1/3} \quad (62)$$

$$\text{Scenario 4: } v_i \sim (\text{Beta}(.5, .5) - 1/2) / \sqrt{1/8} \quad (63)$$

In all scenarios, the exogenous component is scaled to have zero mean and unit variance to enable comprehensive scenarios. We consider two skewed and two symmetric (but leptokurtic) distributions.

		Scenario 1				Scenario 2				Scenario 3				Scenario 4			
		Mean	SD	RMSE	t_{bias}	Mean	SD	RMSE	t_{bias}	Mean	SD	RMSE	t_{bias}	Mean	SD	RMSE	t_{bias}
OLS	$\hat{\alpha}$	2.00	.023	.023	-.019	2.00	.023	.023	.042	1.99	.023	.023	-.052	2.00	.022	.022	-.016
	$\hat{\beta}$	1.49	.023	0.50	21.9	1.51	.061	.516	8.38	1.50	.013	0.50	37.1	1.49	.013	0.50	39.5
Proposed	$\hat{\alpha}$	2.00	.033	.033	-.008	2.00	.032	.032	-.091	2.00	.033	.033	-.092	2.00	.031	.031	-.034
	$\hat{\beta}$	.993	.084	.084	-.089	.997	.052	.054	-.055	.998	.039	.039	-.053	.998	.034	.034	-.048

Table 11: True values are $\alpha = 2$, and $\beta = 1$. For further notes, see Table 1.

Table 11 presents the results. The proposed estimator consistently recovers the true

value of β , with minimal bias and low RMSE across all settings. The results for the intercept are also accurate under the proposed method and largely unaffected by the distribution of v_i . These findings confirm that the effectiveness of our approach does not depend on the specific shape of the non-Gaussian distribution in the exogenous component. Both skewed and symmetric (but leptokurtic) cases are handled robustly.

4.8 Varying the error distribution

Consider the following DGP:

$$y_i = \alpha + \beta p_i + \xi_i, \quad \xi_i = \xi_i^a + \xi_i^b \quad (64)$$

$$p_i = v_i + \xi_i^a, \quad v_i \sim \text{Ga}(1, 1) \quad (65)$$

$$\text{Scenario 1: } \xi_i^a \sim t(4)/\sqrt{2}, \quad \xi_i^b = 0 \quad (66)$$

$$\text{Scenario 2: } \xi_i^a \sim t(4)/\sqrt{2}, \quad \xi_i^b \sim \mathcal{N}(0, 1) \quad (67)$$

$$\text{Scenario 3: } \xi_i^a \sim \text{Unif}[-1, 1]/\sqrt{1/3}, \quad \xi_i^b = 0 \quad (68)$$

$$\text{Scenario 4: } \xi_i^a \sim \text{Unif}[-1, 1]/\sqrt{1/3}, \quad \xi_i^b \sim \mathcal{N}(0, 1) \quad (69)$$

$$\text{Scenario 5: } \xi_i^a \sim (\text{Beta}(.5, .5) - 1/2) / \sqrt{1/8}, \quad \xi_i^b = 0 \quad (70)$$

$$\text{Scenario 6: } \xi_i^a \sim (\text{Beta}(.5, .5) - 1/2) / \sqrt{1/8}, \quad \xi_i^b \sim \mathcal{N}(0, 1). \quad (71)$$

In scenarios 1, 3, and 5, the error is fully nonnormal and so it is included in the endogenous regressor. In scenarios 2, 4 and 6, an additional noise component is included in the error term (but not in the endogenous regressor). This introduces an additional challenge for the proposed approach, since a common normally distributed component is sought.

		Scenario 1				Scenario 2				Scenario 3			
		Mean	SD	RMSE	t_{bias}	Mean	SD	RMSE	t_{bias}	Mean	SD	RMSE	t_{bias}
OLS	$\hat{\alpha}$	1.50	.035	.498	-14.1	1.50	.054	.503	-9.31	1.50	.023	.501	21.6
	$\hat{\beta}$	1.49	.036	.498	13.7	1.49	.042	.499	11.8	1.50	.023	.502	21.8
Proposed	$\hat{\alpha}$	2.00	.076	.076	.014	2.06	.615	.615	.102	2.01	.071	.072	.095
	$\hat{\beta}$	.998	.068	.068	-.027	.934	.617	0.62	.107	.994	.065	.065	-.085
		Scenario 4				Scenario 5				Scenario 6			
		Mean	SD	RMSE	t_{bias}	Mean	SD	RMSE	t_{bias}	Mean	SD	RMSE	t_{bias}
OLS	$\hat{\alpha}$	1.50	.046	.502	-10.9	1.49	.023	.502	-22.3	1.50	.045	.502	-11.0
	$\hat{\beta}$	1.50	.033	.501	15.3	1.50	.023	.502	21.6	1.50	.033	.502	15.3
Proposed	$\hat{\alpha}$	2.06	.121	.134	.474	2.00	.073	.073	0.03	2.07	.127	.147	.582
	$\hat{\beta}$	.941	.114	.128	-.512	.997	.065	.065	-.042	.926	.121	.141	-.061

Table 12: True values are $\alpha = 2$, and $\beta = 1$. For further notes, see Table 1.

Table 13 presents the results. The proposed estimator performs very well in Scenarios 1, 3, and 5. In these settings, the method accurately recovers both the intercept and the slope, with low RMSE and minimal bias. However, performance slightly deteriorates in Scenarios 2, 4, and 6, where only part of the error term overlaps with the regressor. In these cases, the estimator still outperforms OLS but exhibits some bias and increased variance, particularly in Scenario 2, where the combination of a heavy-tailed t -distributed endogenous component and a Gaussian noise term makes identification more difficult. Nevertheless, these results highlight the strength of the proposed approach when the endogeneity arises entirely from a non-Gaussian source shared with the regressor. When additional independent noise is added to the error term, the method is only slightly affected.

4.9 Additional normal noise in the error term

To complete the simulations from the previous subsection, we next consider the scenario of additional normal noise in the error term. Consider the following DGP:

$$y_i = \alpha + \beta p_i + \xi_i, \quad \xi_i = \xi_i^a + \delta \xi_i^b \quad (72)$$

$$p_i = v_i + \xi_i^a, \quad v_i \sim \text{Ga}(1, 1), \quad \xi_i^a \sim \mathcal{N}(0, 1), \quad \xi_i^b \sim \mathcal{N}(0, 1), \quad (73)$$

where $\delta = \{1, 2, 4, 8\}$ controls if the error distribution is dominated by the noise component ξ_i^b . The challenge in this scenario is that the error consists of an additional normal component that is exogenous and thus can lead to confusion.

		$\delta = 1$				$\delta = 2$				$\delta = 4$				$\delta = 8$			
		Mean	SD	RMSE	t_{bias}	Mean	SD	RMSE	t_{bias}	Mean	SD	RMSE	t_{bias}	Mean	SD	RMSE	t_{bias}
OLS	$\hat{\alpha}$	1.50	.046	.502	-10.95	1.50	.084	.504	-5.90	1.50	.153	.523	-3.26	1.51	.311	.579	-1.57
	$\hat{\beta}$	1.50	.033	.501	15.0	1.50	.053	.502	9.35	1.51	.091	.514	5.56	1.50	.184	.531	2.72
Proposed	$\hat{\alpha}$	2.01	.126	.126	.052	2.02	.231	.232	.082	2.00	.938	.938	.007	1.92	1.99	1.99	-.038
	$\hat{\beta}$	.992	.119	.119	-.065	.983	.223	.224	-.077	1.00	.383	.383	.006	1.09	1.93	1.93	.044

Table 13: True values are $\alpha = 2$, and $\beta = 1$. For further notes, see Table 1.

Table 13 presents the simulation results. The proposed method consistently delivers estimates that are both unbiased and come with small variability. Even as δ grows large, i.e., more noise in the error term is added that can be confused with the endogeneity, the point estimates remain close to the true values, and the RMSE reflects the increasing noise without inflating the bias. For example, when $\delta = 1$, the estimator for the slope is nearly unbiased and has a low RMSE of .119. When $\delta = 8$, the RMSE increases, as expected due to the large variance in the error term, but the estimate of $\beta = 1.09$ remains close to the truth. Overall, the results show that the proposed estimator is robust to additive exogenous noise, whereas OLS fails to recover the true structural parameters. However, the standard errors become huge.

4.10 Additional normal noise in the endogenous regressor

Consider the following DGP:

$$y_i = \alpha + \beta p_i + \xi_i, \quad \xi_i \sim \mathcal{N}(0, 1) \quad (74)$$

$$p_i = v_i + \xi_i + \delta r_i, \quad v_i \sim \text{Ga}(1, 1), \quad r_i \sim \mathcal{N}(0, 1) \quad (75)$$

where $\delta = \{1, 1.5, 2, 3\}$ controls if the distribution of the endogenous regressor is dominated by the normal noise component r_i . The challenge in this setting is that the endogenous regressor consists of an exogenous normal component that can be confused with ξ .

		$\delta = 1$				$\delta = 1.5$				$\delta = 2$				$\delta = 3$			
		Mean	SD	RMSE	t_{bias}	Mean	SD	RMSE	t_{bias}	Mean	SD	RMSE	t_{bias}	Mean	SD	RMSE	t_{bias}
OLS	$\hat{\alpha}$	1.67	.029	.334	11.7	1.77	.031	.236	-7.64	1.83	0.03	.168	-5.55	1.91	.032	.097	-2.89
	$\hat{\beta}$	1.33	.017	.335	19.7	1.23	.014	.235	16.6	1.17	.012	.167	14.0	1.09	.009	.092	9.85
Proposed	$\hat{\alpha}$	2.02	.201	.201	.098	1.97	2.09	2.09	-.014	2.01	4.84	4.83	.002	1.90	21.8	21.8	-.005
	$\hat{\beta}$	0.98	.209	0.21	-.097	1.03	2.18	2.18	.013	.989	4.76	4.76	-.022	1.11	22.8	22.8	.005

Table 14: True values are $\alpha = 2$, and $\beta = 1$. For further notes, see Table 1.

Table 14 presents the results for different values of δ . The proposed method consistently recovers estimates for β that are close to the true value of 1, regardless of the value of δ . However, in the case of $\delta = 3$, a bias arises of similar size of that of the OLS estimator. The standard errors are strongly influenced. While these are still moderate in the case of $\delta = 1$, they increase exorbitantly for larger delta values. In case of $\delta = 3$, the standard error for β is more than 2,500 times higher than that of the OLS estimator. In empirical applications, exorbitantly high standard errors can flag deviations for model identification.

5 Discussion

Many studies struggle with endogeneity where the regressor P is correlated with the unobserved error in the outcome Y because of a missing factor. The usual fix is an instrument,

but good instruments are hard to find. A popular workaround, the Gaussian-copula control function (GCCF), was thought to solve this without instruments. Recent work, however, shows GCCF actually makes things worse when the endogeneity comes from a plain, additive omitted variable—the most common situation in marketing data (Dost and Haschka, 2025).

To remedy that gap, the authors reconceptualise the bias as a blind-source-separation task: the observed regressor and outcome are mixtures of (i) a latent normal confounder and (ii) an independent, non-normal exogenous shock to the regressor. Exploiting the statistical independence and the fact that at least one component is non-Gaussian, they apply Independent Component Analysis (ICA) to recover the latent confounder. Inserting this recovered factor as a control function then yields consistent estimates of the causal effect. The approach assumes: 1. The true error in Y is normally distributed. 2. The hidden confounder that drives both P and Y is also normal. 3. The part of P that isn't from the confounder (an exogenous shock) is not normal; if P as a whole is not normal can be checked in the data.

Because one hidden source is normal and the other is not, Independent Component Analysis (ICA) can pull them apart. Once the method extracts the normal confounder, the authors add that piece as a control in the outcome regression, purging the bias and giving a clean estimate of the causal effect of P on Y .

Extensive Monte-Carlo evidence underscores three take-aways:

- Handling of endogeneity because of omitted variables. We explicitly simulate scenarios with up to 20 omitted variables of different distributions and show that the estimator is unbiased.
- Breadth of performance. Across a wide menu of data-generating processes—including ones tailor-made for GCCF—the new estimator matches or beats GCCF, LIV, and 2SLS in bias and efficiency.
- Small-sample robustness. It works well with small samples ($n=250$ or $n=500$), the sizes many field studies have.

- Graceful degradation. It only breaks down when its own identifying conditions are strongly violated (e.g., the latent confounder becomes highly non-Gaussian or the exogenous component drifts toward normality); otherwise it still approximates the truth.

Practically, the method gives researchers an “instrument-free IV” that is purpose-built for the most common confounding structure they face, yet requires little more than checking the regressor’s distribution and running the supplied ICA routine. An open-source R routine automates the ICA step, adds the recovered control factor, and supplies bootstrap standard errors, so running the method is about as easy as a regular two-step regression. Limitations include sensitivity to severe departures from normality in the confounder and potential complexity if multiple latent confounders are present—fertile ground for future extensions (e.g., over-complete ICA or semi-parametric relaxations).

In sum, we deliver a theoretically sound, easy-to-implement alternative that restores credibility to causal claims in precisely the scenarios where the reigning Gaussian-copula fix falters.

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Appendix

Fancy appendix

Acknowledgements

Thanks to all humans for their contribution.